Dictionary

```
It is collection of keys and values.
```

Properties

In [ ]:
```
1)Mutable
2)Ordered/Unordered
3)We can not use indexing and slicing.
4)Key can not be duplicate
5)Values can be duplicates.
6)Enclosed by {}
7)Initialize dictionary by using dict() or {}

Python2->Unordered
Python 3->Ordered

Data Types of key:-Immutable(int,float,complex,str,tuple,)

Data Types of value:-Any Data type(int,float,complex,str,list,tuple,set,dict

    dict1=
```

Syntax:

In [ ]:
```
dict1={key:Value}

dict2={}=dict()  #dictionary initization
```

In [2]:
```
d1={"a":97,"b":98,"c":122}
d1
```

Out[2]: {'a': 97, 'b': 98, 'c': 122}

In [3]:
```
d1={"a":97,"b":98,"c":122}
d1["a"]
```

Out[3]: 97

In [4]:
```
d1={"a":97,"b":98,"c":122}
d1["B"]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[4], line 2
      1 d1={"a":97,"b":98,"c":122}
----> 2 d1["B"]

KeyError: 'B'
```

In [5]:
```
d2={"a":33,"b":22,"c":[111,222,333]}
d2
```

Out[5]: {'a': 33, 'b': 22, 'c': [111, 222, 333]}

In [6]:
```
d2={"a":33,"b":22,"c":{111,222,333}}
d2
```

Out[6]: {'a': 33, 'b': 22, 'c': {111, 222, 333}}

In [7]:
```
len(d2)
```

Out[7]: 3

Dictionary Functions

1.get()

In [ ]:
```
dict.get("key")
```

In [9]:
```
dict1={'A':100,'B':200,'C':300,'D':400}
dict1['A']
```

Out[9]: 100

In [11]:
```
dict1={'A':100,'B':200,'C':300,'D':400}
dict1.get('A')
```

Out[11]: 100

2)keys()

In [ ]:
```
dict.keys()
```

In [12]:
```
dict1={'A':100,'B':200,'C':300,'D':400}
dict1.keys()
```

Out[12]:
```
dict_keys(['A', 'B', 'C', 'D'])
```

In [13]:
```
dict1={'A':100,'B':200,'C':300,'D':400}
for i in dict1.keys():
    print(i)
```

```
A
B
C
D
```

In [15]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
len(employee_data)
```

Out[15]:
```
4
```

In [16]:
```
employee_data.get('Designation')
```

Out[16]:
```
'Data Scientist'
```

In [17]:
```
employee_data.keys()
```

Out[17]:
```
dict_keys(['Name', 'Age', 'Designation', 'Salary'])
```

In [20]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
for key in employee_data.keys():
    print("key is:",key)
    print("Values is",employee_data[key])
    print('*'*40)
```

```
key is: Name
Values is Yash
****************************************
key is: Age
Values is 22
****************************************
key is: Designation
Values is Data Scientist
****************************************
key is: Salary
Values is 70000
****************************************
```

3)values()

In [ ]:
```
Synatx:Dict.values()
```

In [21]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
employee_data.values()
```

Out[21]:
```
dict_values(['Yash', 22, 'Data Scientist', 70000])
```

In [22]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
for value in employee_data.values():
    print("Values is:",value)
    print("*"*40)
```

```
Values is: Yash
****************************************
Values is: 22
****************************************
Values is: Data Scientist
****************************************
Values is: 70000
****************************************
```

4)items()

In [ ]:
```
Synatx: Dict.items()
```

In [23]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
employee_data.items()
```

Out[23]:
```
dict_items([('Name', 'Yash'), ('Age', 22), ('Designation', 'Data Scientis
t'), ('Salary', 70000)])
```

In [24]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
for i in employee_data.items():
    print(i)
```

```
('Name', 'Yash')
('Age', 22)
('Designation', 'Data Scientist')
('Salary', 70000)
```

In [26]:
```
a,b=('Name', 'Yash')
a
```

Out[26]:
```
'Name'
```

In [27]:
```
b
```

Out[27]:
```
'Yash'
```

In [28]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
for key,value in employee_data.items():
    print(f"key=={key} and value=={value}")
    print("*"*40)
```

```
key==Name and value==Yash
****************************************
key==Age and value==22
****************************************
key==Designation and value==Data Scientist
****************************************
key==Salary and value==70000
****************************************
```

5)Update

In [ ]:
```
Update method Inserts the specified items to the dictionary
```

In [1]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
employee_data
```

Out[1]:
```
{'Name': 'Yash', 'Age': 22, 'Designation': 'Data Scientist', 'Salary': 700
00}
```

In [2]:
```
employee_data['Age']
```

Out[2]:
```
22
```

In [3]:
```
employee_data['Age']=32
employee_data
```

Out[3]:
```
{'Name': 'Yash', 'Age': 32, 'Designation': 'Data Scientist', 'Salary': 700
00}
```

In [5]:
```
employee_data.update({"Salary":80000})
employee_data
```

Out[5]:
```
{'Name': 'Yash', 'Age': 32, 'Designation': 'Data Scientist', 'Salary': 800
00}
```

In [7]:
```
employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary
employee_data.update({"Location":"Pune"})
employee_data
```

Out[7]:
```
{'Name': 'Yash',
 'Age': 22,
 'Designation': 'Data Scientist',
 'Salary': 70000,
 'Location': 'Pune'}
```

In [10]:
```
list1=[2,3,4,5,6,7,8]
#square_dict1={2:4,3:9,4:16,5:25,6:36,7:49,8:64}
square_dict1={}
for i in list1:
    square=i**2
    square_dict1.update({i:square})
print(square_dict1)
```

```
{2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}
```

In [11]:
```
A={}
A.update({'A':100})
A.update({'B':200})
A
```

Out[11]:
```
{'A': 100, 'B': 200}
```

In [12]:
```
square=dict()
for i in range(11,21):
    s=i**2
    square.update({i:s})
square
```

Out[12]:
```
{11: 121,
 12: 144,
 13: 169,
 14: 196,
 15: 225,
 16: 256,
 17: 289,
 18: 324,
 19: 361,
 20: 400}
```

ASCII

In [ ]:
```
{'A':65,'B':66......'Z':90}
```

In [16]:
```
ascii={}
for i in range(65,91):
    #print(i,chr(i))
    ascii.update({chr(i):i})
ascii
```

Out[16]:
```
{'A': 65,
 'B': 66,
 'C': 67,
 'D': 68,
 'E': 69,
 'F': 70,
 'G': 71,
 'H': 72,
 'I': 73,
 'J': 74,
 'K': 75,
 'L': 76,
 'M': 77,
 'N': 78,
 'O': 79,
 'P': 80,
 'Q': 81,
 'R': 82,
 'S': 83,
 'T': 84,
 'U': 85,
 'V': 86,
 'W': 87,
 'X': 88,
 'Y': 89,
 'Z': 90}
```

Delete Items From Dictionary

In [ ]:
```
1)pop()
2)popitem
3)clear
```

1)pop()

In [ ]:
```
Dict.pop(key)
```

In [17]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.pop('B')
data
```

Out[17]:
```
{'A': 100, 'C': 300, 'D': 400, 'E': 500}
```

In [19]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.pop('B')
data.pop('A')
data
```

Out[19]:
```
{'C': 300, 'D': 400, 'E': 500}
```

2)popitem()

In [ ]:
```
Dict.popitem()

It will delete last inserted item (key and value pair)
```

In [20]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.popitem()
data
```

Out[20]:
```
{'A': 100, 'B': 200, 'C': 300, 'D': 400}
```

In [21]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.popitem()
data.popitem()
data
```

Out[21]:
```
{'A': 100, 'B': 200, 'C': 300}
```

3)clear()

In [ ]:
```
Dict.clear()

It will delete all items from dictionary

It return Null dictionary.
```

In [22]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.clear()
data
```

Out[22]:
```
{}
```

In [23]:
```
data
```

Out[23]:
```
{}
```

4. Del keyword

In [24]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
data['C']
```

Out[24]:
```
300
```

In [27]:
```
data['C']=350
data
```

Out[27]:
```
{'A': 100, 'B': 200, 'C': 350, 'D': 400, 'E': 500}
```

In [29]:
```
del data['C']
data
```

```
---------------------------------------------------------------------------
-
KeyError                                  Traceback (most recent call las
t)
Cell In[29], line 1
----> 1 del data['C']
      2 data

KeyError: 'C'
```

In [31]:
```
data={'A':100,'B':200,'C':300,'D':400,'E':500}
print(data)
del data
print(data)
```

```
{'A': 100, 'B': 200, 'C': 300, 'D': 400, 'E': 500}

---------------------------------------------------------------------------
-
NameError                                 Traceback (most recent call las
t)
Cell In[31], line 4
      2 print(data)
      3 del data
----> 4 print(data)

NameError: name 'data' is not defined
```

fromkeys

```
Dict.fromkeys(iterable,[Value])

Create dictionary with same value.

Default value is None.
```

In [32]:
```
x=list("ASDFG")
x
```

Out[32]:
```
['A', 'S', 'D', 'F', 'G']
```

In [33]:
```
x=list("Python")
x
```

Out[33]:
```
['P', 'y', 't', 'h', 'o', 'n']
```

In [34]:
```
new_dict=dict.fromkeys(x)
new_dict
```

Out[34]:
```
{'P': None, 'y': None, 't': None, 'h': None, 'o': None, 'n': None}
```

In [35]:
```
new_dict=dict.fromkeys('Python',500)
new_dict
```

Out[35]:
```
{'P': 500, 'y': 500, 't': 500, 'h': 500, 'o': 500, 'n': 500}
```

In [36]:
```
new_dict=dict.fromkeys((2,3,4,5,6),500)
new_dict
```

Out[36]:
```
{2: 500, 3: 500, 4: 500, 5: 500, 6: 500}
```

In [37]:
```
new_dict=dict.fromkeys('Python',[10,20,30,40,50])
new_dict
```

Out[37]:
```
{'P': [10, 20, 30, 40, 50],
 'y': [10, 20, 30, 40, 50],
 't': [10, 20, 30, 40, 50],
 'h': [10, 20, 30, 40, 50],
 'o': [10, 20, 30, 40, 50],
 'n': [10, 20, 30, 40, 50]}
```

In [38]:
```
new_dict=dict.fromkeys('Python','Data science')
new_dict
```

Out[38]:
```
{'P': 'Data science',
 'y': 'Data science',
 't': 'Data science',
 'h': 'Data science',
 'o': 'Data science',
 'n': 'Data science'}
```

In [42]:
```
sub_name=["Bio","Chem","Hindi","Math","Eng",'Phy']
marks=[88,90,76,65,87,80]
result={}
for i,value in enumerate(sub_name):
    result.update({value:marks[i]})
    print(result)
```

```
{'Bio': 88}
{'Bio': 88, 'Chem': 90}
{'Bio': 88, 'Chem': 90, 'Hindi': 76}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
```

setdefault()

In [ ]:
```
Dict.setdefault(key,[Val])

Default value is None.
```

In [43]:
```
result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80
result['Chem']
```

Out[43]: 90

In [44]:
```
result.get('Chem')
```

Out[44]: 90

In [46]:
```
result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80
result.setdefault('Chem')
result
```

Out[46]: {'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

In [50]:
```
result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80
result.update({'Chem':100})
result
```

Out[50]: {'Bio': 88, 'Chem': 100, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

In [51]:
```
result.setdefault("Chem",67)
result
```

Out[51]: {'Bio': 88, 'Chem': 100, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

In [52]:
```
result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80
result.setdefault('Python')
result
```

Out[52]:
```
{'Bio': 88,
 'Chem': 90,
 'Hindi': 76,
 'Math': 65,
 'Eng': 87,
 'Phy': 80,
 'Python': None}
```

Copy()

In [1]:
```
employee_data={"Name":"Neha",
              "Age":22,
              "Company Name":"Wipro"}
new_dict=employee_data

new_dict.update({"Salary":70000})

print(employee_data)
print(new_dict)
```

```
{'Name': 'Neha', 'Age': 22, 'Company Name': 'Wipro', 'Salary': 70000}
{'Name': 'Neha', 'Age': 22, 'Company Name': 'Wipro', 'Salary': 70000}
```

In [3]:
```
employee_data={"Name":"Neha",
              "Age":22,
              "Company Name":"Wipro"}
new_dict=employee_data.copy()

new_dict.update({"Salary":60000})

print(employee_data)
print(new_dict)
```

```
{'Name': 'Neha', 'Age': 22, 'Company Name': 'Wipro'}
{'Name': 'Neha', 'Age': 22, 'Company Name': 'Wipro', 'Salary': 60000}
```

Nested Dictionary

In [ ]:
```
Dictionary as a value in dictionary
```

In [4]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'}}
len(employee_data)
```

Out[4]:
```
1
```

In [5]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'}}
employee_data
```

Out[5]:
```
{'TCS10001': {'Name': 'Prerana',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Scientist'}}
```

In [6]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'}}
employee_data
```

Out[6]:
```
{'TCS10001': {'Name': 'Prerana',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Scientist'},
 'TCS10002': {'Name': 'Neha',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Python developer'},
 'TCS10003': {'Name': 'Manjusha',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Analyst'},
 'TCS10004': {'Name': 'Yash',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Scientist'}}
```

In [7]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'}}
employee_data["TCS10001"]
```

Out[7]:
```
{'Name': 'Prerana',
 'Age': 22,
 'Company_name': 'TCS',
 'Designation': 'Data Scientist'}
```

In [8]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'}}
employee_data["TCS10001"]['Designation']
```

Out[8]: 'Data Scientist'

In [10]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'}}
employee_data.get('TCS10001')['Designation']
```

Out[10]: 'Data Scientist'

In [11]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist',
                          "Salary":[20000,30000,40000,50000]}}
employee_data['TCS10004']['Salary'][-1]
```

Out[11]:
```
50000
```

In [13]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                          "Age":22,
                          'Company_name':"TCS",
                          "Designation":'Data Scientist',
                          "Salary":[20000,30000,40000,50000]}}
employee_data['TCS10004']['Salary'].append(60000)
employee_data
```

Out[13]:
```
{'TCS10001': {'Name': 'Prerana',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Scientist'},
 'TCS10002': {'Name': 'Neha',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Python developer'},
 'TCS10003': {'Name': 'Manjusha',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Analyst'},
 'TCS10004': {'Name': 'Yash',
  'Age': 22,
  'Company_name': 'TCS',
  'Designation': 'Data Scientist',
  'Salary': [20000, 30000, 40000, 50000, 60000]}}
```

In [14]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist',
                           "Salary":[20000,30000,40000,50000]}}
employee_data['TCS10004']['Designation'][5:]
```

Out[14]: 'Scientist'

In [15]:
```
employee_data={'TCS10001':{"Name":"Prerana",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist'},
              'TCS10002':{"Name":"Neha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Python developer'},
              'TCS10003':{"Name":"Manjusha",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Analyst'},
               'TCS10004':{"Name":"Yash",
                           "Age":22,
                           'Company_name':"TCS",
                           "Designation":'Data Scientist',
                           "Salary":[20000,30000,40000,50000]}}
employee_data['TCS10004']['Designation'].split()[-1]
```

Out[15]: 'Scientist'

Dictionary Comprehension

In [17]:
```
list1=[3,4,5,6,7]
sq=[i**2 for i in list1]
sq
```

Out[17]: [9, 16, 25, 36, 49]

In [18]:
```
list1=[3,4,5,6,7]
sq=[i for i,value in enumerate(list1)]
sq
```

Out[18]: [0, 1, 2, 3, 4]

In [19]:
```
list1=[3,4,5,6,7]
sq=[value**2 for i,value in enumerate(list1)]
sq
```

Out[19]:
```
[9, 16, 25, 36, 49]
```

In [20]:
```
cube={}
for i in range(1,11):
    cube.update({i:i**3})
    print(cube)
```

```
{1: 1}
{1: 1, 2: 8}
{1: 1, 2: 8, 3: 27}
{1: 1, 2: 8, 3: 27, 4: 64}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512, 9: 729}
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512, 9: 729, 10: 100
0}
```

In [21]:
```
cube={}
for i in range(1,11):
    if i%2==0:
        cube.update({i:i**3})
        print(cube)
```

```
{2: 8}
{2: 8, 4: 64}
{2: 8, 4: 64, 6: 216}
{2: 8, 4: 64, 6: 216, 8: 512}
{2: 8, 4: 64, 6: 216, 8: 512, 10: 1000}
```

In [22]:
```
cube={i:i**3 for i in range(1,11)}
print(cube)
```

```
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512, 9: 729, 10: 100
0}
```

In [24]:
```
cube={i:i**3 for i in range(1,11) if i%2==0}
cube
```

Out[24]:
```
{2: 8, 4: 64, 6: 216, 8: 512, 10: 1000}
```

In [26]:
```
name="Python"
name[0]="J"
print(name)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[26], line 2
      1 name="Python"
----> 2 name[0]="J"
      3 print(name)

TypeError: 'str' object does not support item assignment
```

In [ ]: